

Part I: Research Question

A.

When it comes to strong and healthy bones, we are taught from a young age that calcium is the way to go. What we do not necessarily learn however, is that you need sufficient amounts of vitamin D to help your body absorb that calcium efficiently (Meeks (MS, RD) & Devje (RD CDE MSc), 2023) (Bilezikian (MD), 2022)). Vitamin D deficiency can lead to multiple health problems, but an overdose of vitamin D is also dangerous.

With this medical data set, we can deep dive into patient history to see if their demographic features (location, education, etc.) and how their nutritional habits (vitamin supplements, etc.) affect their vitamin D levels. We can also investigate other diseases or issues such as diabetes and being overweight, to see if it is connected to vitamin D levels. It is important to understand how these aspects influence one another and if they are connected so that satisfactory care and better guidance can be given to patients.

B.

	Variable	Description	Type	Format	Examples
1	Case Order	Placeholder for dataset order	Qualitative	Integer	1,2,3...
2	Customer ID	Unique identification	Qualitative	Object	C412403
3	Interaction	Unique identification	Qualitative	Object	8cd49b13-f45a-4b47-a2bd-173ffa932c2f
4	UID	Unique identification	Qualitative	Object	3a83ddb66e2ae73798bdf1d705dc0932
5	City	City where patient lives	Qualitative	Object	Nantucket, Albany, Daleville
6	State	State where patient lives	Qualitative	Object	MA, NY, VA
7	County	County where patient lives	Qualitative	Object	Ottawa, Morgan, Waseca
8	Zip	Zip code where patient lives	Qualitative	Integer	51250, 35811, 92225
9	Lat (Latitude)	GPS Coordinates	Qualitative	Float	40.38836
10	Lng (Longitude)	GPS Coordinates	Qualitative	Float	-102.10258
11	Population	Number of people within 1 mile radius of patient	Quantitative	Integer	13947, 729, 12
12	Area	Rural/Urban/Suburban	Qualitative	Object	Urban, Rural, Suburban
13	Time zone	Time zone where patient lives	Qualitative	Object	America/Indiana/Indianapolis
14	Job	Patient's occupation	Qualitative	Object	Horticulturist, commercial, Contractor, Restaurant manager, fast food
15	Children	Household's number of children	Quantitative	Float	1, 3, NA
16	Age	Patient's age	Quantitative	Float	18, 56, 25
17	Education	Highest level of education	Qualitative	Object	Regular High School Diploma, Master's Degree
18	Employment	Status of employment	Qualitative	Object	Full Time, Retired, Unemployed
19	Income	Annual income	Quantitative	Float	55586.48
20	Marital	Marital Status	Qualitative	Object	Married, Widowed

21	Gender	Patient's gender	Qualitative	Object	Male, Female
22	ReAdmins (Re-admissions)	Was patient readmitted within a month after previous release	Qualitative	Object	Yes, No
23	Vit D Levels	ng/mL measure vitamin D levels	Quantitative	Float	16.49455
24	Doc Visits	Number of times doctor visited patient	Quantitative	Integer	4, 7, 2...
25	Full Meals Eaten	Patient's number of full meals during stay	Quantitative	Integer	0, 1, 2...
26	Vit D Supp (Supplement)	Number of vitamin D supplements given during stay	Quantitative	Integer	0, 1, 2...
27	Soft Drink	Does patient drink more than 3 sodas habitually	Qualitative	Object	Yes, No, NA
28	Initial Admin (Admission)	Admission type	Qualitative	Object	Emergency Admission, Observation Admission
29	High Blood	Patient's blood pressure	Qualitative	Object	Yes, No
30	Stroke	Previous strokes	Qualitative	Object	Yes, No
31	Complication Risk	Patient's level of complication risk	Qualitative	Object	Low, medium
32	Overweight	Yes or no for specific condition/disease	Qualitative	Float	0, 1, NA
33	Arthritis	Yes or no for specific condition/disease	Qualitative	Object	Yes, No
34	Diabetes	Yes or no for specific condition/disease	Qualitative	Object	Yes, No
35	Hyperlipidemia	Yes or no for specific condition/disease	Qualitative	Object	Yes, No
36	Backpain	Yes or no for specific condition/disease	Qualitative	Object	Yes, No
37	Anxiety	Yes or no for specific condition/disease	Qualitative	Float	0, 1, NA
38	Allergic Rhinitis	Yes or no for specific condition/disease	Qualitative	Object	Yes, No
39	Reflux Esophagitis	Yes or no for specific condition/disease	Qualitative	Object	Yes, No
40	Asthma	Yes or no for specific condition/disease	Qualitative	Object	Yes, No
41	Services	Type of services received during stay	Qualitative	Object	Intravenous, Blood Work
42	Initial Days	Number of days in hospital for first visit	Quantitative	Float	5.95725, 1.012586
43	Total Charge	Daily costs of patient's stay	Quantitative	Float	2990.086, 3316.4
44	Additional Charges	Other charges during stay	Quantitative	Float	18518.04, 10575.43
45	Item 1: Timely admission	Survey question	Qualitative	Integer	1,2,3,4...
46	Item 2: Timely treatment	Survey question	Qualitative	Integer	1,2,3,4...

47	Item 3: Timely visits	Survey question	Qualitative	Integer	1,2,3,4...
48	Item 4: Reliability	Survey question	Qualitative	Integer	1,2,3,4...
49	Item 5: Options	Survey question	Qualitative	Integer	1,2,3,4...
50	Item 6: Hours of treatment	Survey question	Qualitative	Integer	1,2,3,4...
51	Item 7: Courteous staff	Survey question	Qualitative	Integer	1,2,3,4...
52	Item 8: Evidence of active listening from doctor	Survey question	Qualitative	Integer	1,2,3,4...

Part II: Data-Cleaning Plan

C.

1.

Step 1: Import the csv file into the Python shell using `pd.read_csv()`. Print the first few rows of the data frame using `head()` to ensure that the import was successful.

Step 2: Review the dataset to see if there are any duplicate entries by calling `duplicated().sum()` on the data frame – this will give the total number of duplicated rows. Remove any duplicated rows.

Step 3: Investigate the formatting of data types for each column, by calling `info()` on the data frame. This will list the columns, count of non-null values and their data type. Reformat the data types where necessary using `astype()` on each column.

Step 4: Compare the entries in the data frame with the information in the data dictionary to ensure consistency between the two sources and the context. There is no function or module to call for this – it will need to be done manually. To change information in an entry, use the `replace()` function.

Step 5: Check for null values by calling the `isna().sum()` function on the data frame to count the number of missing rows in each column. Dividing each column by the total number of rows using the `len()` function, it displays the percentage of missing values in each column.

Step 6: Identify the type of missingness of the data to see if the values are missing at random or not. By creating a missingness matrix (`msno.matrix`) and ordering the results by different columns (with missing values) the type of missingness can be determined. Find the right methods to handle the missing values based on the context – either drop, fill with zero's or impute. Fill null values using the `fillna()` function, or by imputing values through the k-nearest neighbour method, averages or most frequent entries.

Step 7: The outliers need to be identified. First, the non-numerical columns need to be removed so only the numerical columns remain. By calling `describe()` on the smaller data frame, statistical data for quantitative columns are displayed. Using `sns.histplot`, histograms of the qualitative data are displayed to explore the distribution of the entries. When data is normally distributed the z-score is calculated to determine the outliers. When data is not normally distributed, the inter quartile range is calculated to determine the outliers. Find the right methods to handle the outliers based on the context – either drop or change.

2.

Step 1: No mitigation necessary.

Step 2: Duplicate entries do not add any value to the data since it just repeats what is already in another entry. Some issues that can be caused by duplicate entries include skewed data, misleading predictions, and longer processing times.

Step 3: When assessing qualitative data, it is important to verify what data type the entries were imported as. Python automatically assigns it as a numerical data type when numbers are involved, because it does not know the context of the data. Since statistical or mathematical calculations are only done on quantitative columns, changing the data type will ensure that the numerical qualitative columns are excluded. When assessing quantitative data, it is important to validate the range of your data to ensure that the numbers make sense.

Step 4: Consistency and context play an important role in data analysis. By comparing entries in the data set to the data dictionary ensures that the information makes sense, and the entries are what is expected.

Step 5: Null values can have a major influence on the data when not handled correctly. In some cases, it is better to drop null values because imputation does not make sense, in some cases it is better to leave the null values untouched because a value cannot be inferred based on the rest of the data, and in some cases, it makes sense to impute missing values based on the rest of the data (using the mean or zero, etc).

Step 6: By knowing the type of missingness, it assists in choosing the right methods for handling the missing values.

Step 7: Just like null values, outliers can greatly affect the outcome of data analysis. Identifying the outliers and understanding the context of why it is present, assists in choosing the right methods for handling the outliers.

3.

The program language Python was used to write the data cleaning code. Python is a very user-friendly language that integrates with its packages and libraries (I find) in a more understandable way than R. The programming syntax of Python is easy to follow and understand, and changing or updating the data frame in Python is simple.

In some areas, the flow and steps or naming conventions in Python make more sense or are easier to follow. In some areas R makes it easier for the user to follow the flow. My personal preference for this assignment was Python since I understand it better and had an easier time getting used to the syntax. I was struggling with grasping the syntax in R.

Package/Module/Library	Description	Comparison
pandas	Data analysis and manipulation tool.	Dplyr is very similar, just some syntax differences.
missingno	Visualize and analyse missing data (does not need matplotlib to visualize data).	Visdat (needs ggplot to visualize the data).
matplotlib	Visualization tool (static & dynamic)	GGplot
seaborn	Data visualization tool based on matplotlib.	
numpy	Library used for working with arrays.	Matrix package.
Scikit learn	Machine learning module that includes classes and functions for imputation, clustering, feature selection, regression, etc.	kmeans clustering is already available in the base R program. Other packages need to be installed just like Python.
KNN Imputer (imputer)	To perform k-nearest neighbour imputation	
Simple Imputer (imputer)	To perform simple imputation using mean, median or most frequent	
Standard Scaler (preprocessing)	This is used to standardize features by scaling to unit variance after removing the mean	
PCA (Decomposition)	Principal component analysis tool	
Pipeline (Pipeline)	Cross-validate multiple steps together setting different parameters	

In the Python environment the scikit learn package is well known for machine learning processes. In the R environment, on the other hand, needs multiple different packages to perform machine learning processes – like tidymodels, caret & MLR.

4.

See code attached (LorraineF_D206_DataCleaningPlan.py).

Part III: Data Cleaning

D.

1.

Step 1: No issues came up.

Step 2: There were no duplicate values.

Step 3: The following columns were formatted incorrectly.

Numerical columns with qualitative values: Case Order, Zip code, Longitude, and Latitude (should be strings/characters/objects).

Categorical columns (limited by a list of possible entries): Area, Time zone, Employment, Marital, Gender, Initial admission, and Services.

Age, Overweight, and Asthma: formatted as float due to the null values present (should be integer/category)

Step 4: The medical conditions must be coded as “Yes” or “No” – the Overweight and Anxiety columns have 1’s and 0’s.

Step 5: There were multiple columns with missing values (percentage of missing values in parentheses). Children (25.88%), Age (24.14%), Income (24.64%), Soft drink (24.67%), Overweight (9.82%), Anxiety (9.84%), and Initial days (10.56%).

Step 6: The missing values are missing at random – there is no relationship or correlation between the missing values.

Step 7: There were multiple columns with outliers (number of outliers in parentheses). Population (855), Children (303), Income (252), Vitamin D levels (534), Full meals eaten (8), Vitamin D supplements (70), Total charge (466), and Additional charges (424), Item 1 (449), Item 2 (429), Item 3 (443), Item 4 (450), Item 5 (443), Item 6 (443), Item 7 (438), Item 8 (442).

2.

Step 1: Using the correct import method ensures that a data frame is created with the data set entries. Data frames are the right format for data analysis processes.

Step 2: If duplicate values were present, it would make sense to remove them since one entry per patient ID is required.

Step 3: Identifying the correct data type for each column is crucial since each data type has its own methods, attributes and functionalities. It also reduces the memory required to store and run the program which improves the processing time.

Step 4: Re-expressing the 0’s and 1’s as ‘No’ and ‘Yes’ makes the dataset more consistent and reduces the possibility of using the columns for mathematical calculations.

Step 5: Missing values influence the outcome of statistical and mathematical operations which is why it is so important to identify and handle them correctly. Changing or dropping missing values incorrectly might cause biased results in data analysis. By using the KNN (k-nearest neighbour) imputer, Python can use the given data to determine which values are closest to the null values and replace the null values with the imputed value.

Step 6: Identifying the type of missingness helps to determine the steps required for handling missing values. Filling the missing values ensures that all the rows are kept, and no important information gets dropped that might influence the outcome of the research question.

Step 7: Outliers might have negative effects on the results of data analysis, or it might be the information needed to answer a question. Exploring the outliers and handling them correctly will increase the accuracy of your results. Determining why the outliers are there in the first place helps to determine the steps required for handling the outliers. Of all the columns with outliers, we are mostly interested in the Vitamin D levels since this connects to our research question. The outliers in this case are people with vitamin D levels above 40 ng/mol. It is not necessary to get rid of the outliers since we can use them to investigate why the levels might be so high – too much sun exposure, or too many Vitamin D supplements.

3.

Step 1: The file imported successfully.

Step 2: Since no duplicated entries were present, no rows needed to be removed.

Step 3: The columns mentioned above were formatted to the correct data types. The data types for Overweight & Anxiety reformatted as strings once the entries were updated to 'Yes' & 'No'. The Age column data type could only be reformatted to integer once the null values were handles.

Step 4: The Overweight & Anxiety column rows with 0's and 1's was replaced with 'Yes' and 'No' to follow the same formatting as the other medical issue columns.

Step 5: Some columns had a large number of missing values – dropping these rows would reduce the data set significantly and important information might be lost. Since the missing values were missing at random, dropping the rows from one column did not guarantee that there were missing values being dropped in other columns.

Step 6: The missing values in Children, Soft drink, Overweight, and Anxiety were filled with 0's or 'No's - working under the assumption that no entry in the columns meant the patient did not have children, did not drink more than 3 soft drinks on a daily basis, and was not overweight or suffering from anxiety. The Age, Income and Initial Days columns made use of the k-nearest neighbour method to fill missing values with entries that were similar to other entries based on the given information.

Step 7: The Population, Income, Total charge and Additional charge columns had outliers because of the wide range of possible data points. The Children, Full meal eaten, Vitamin D supplements and Survey questions (Item #) had outliers because there were patients with out of the ordinary entries – plausible, but not common.

Based on this information, the outliers were not changed or dropped from the dataset since it gives important insight into the research question.

4.

See attached code (LorraineF_D206_DataCleaning.py).

5.

See attached CSV (LorraineF_D206_CleanedData.csv).

6.

The data cleaning process is limited by not knowing the context of the data points. A missing data point could be attributed to a variety of reasons – it represents a null, it was a typing error, or it was left out on purpose. The outliers in the medical dataset are caused by the variety in the data points – small towns with hundreds of people exist along with cities that have millions of people. The same goes for salary or the number of children. Choosing the incorrect cleaning methods might lead to bad decision making or it could skew the data and further processing won't be accurate. This means the cleaning process is limited by the person handling the data.

Without knowing the context of why some patient's Vitamin D levels are so high, it's possible that the wrong process for handling the data might be used. How the outliers are handled will greatly affect the outcome of the question – remove the outliers and you might miss something important about patients taking too many supplements or you won't be able to find out why their Vitamin D levels are so high.

7.

The research question focuses on investigating a patient's demographic information, health, and wellness and how it affects their vitamin D level as well as how the vitamin D level might influence these things in return. The limitations mentioned above might cause misguided conclusions which could lead to improper care for the patient.

E.

1.

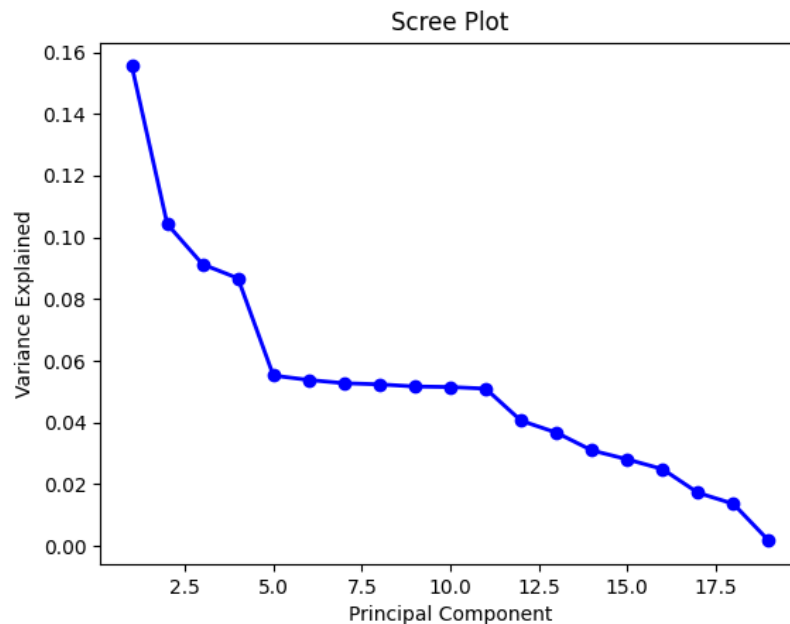
There are 19 principal components in this dataset.

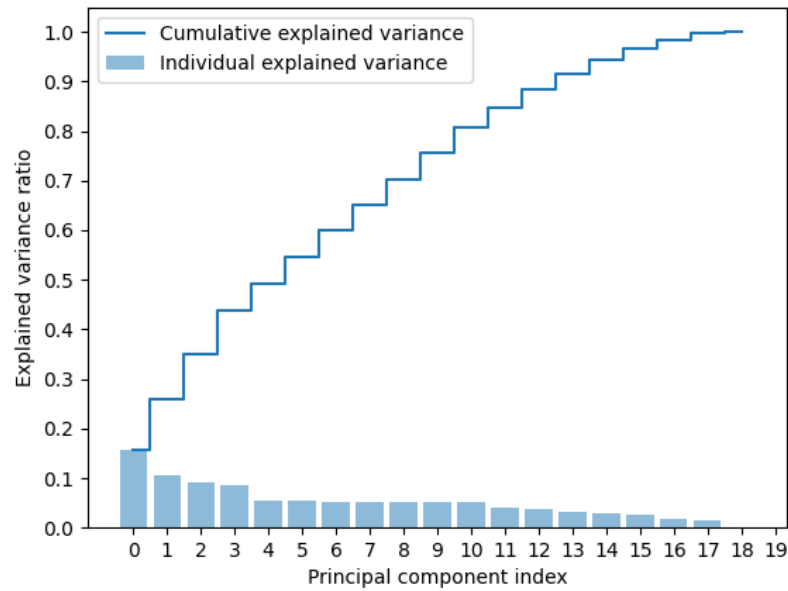
	PC1	PC2	PC3	...	PC17	PC18	PC19
variable				...			
Population	0.010198	0.015008	-0.030072	...	-0.004013	-0.015274	0.000230
Children	0.002294	-0.000616	0.012396	...	-0.001437	-0.005039	-0.001393
Age	0.004036	0.126467	0.694505	...	0.014255	-0.706058	-0.018830
Income	-0.002725	-0.011428	-0.005710	...	-0.006079	-0.011728	0.000749
VitD_levels	-0.009733	0.521811	-0.088268	...	0.010578	-0.016247	0.531079
Doc_visits	0.007145	-0.002163	0.017804	...	-0.001685	0.003011	-0.000774
Full_meals_eaten	-0.000622	-0.007207	0.032611	...	0.001428	-0.007125	-0.002475
VitD_supp	-0.005027	0.029765	0.011274	...	-0.000108	-0.000667	0.001900
Initial_days	-0.021419	0.453259	-0.097979	...	-0.017434	-0.015810	0.465276
TotalCharge	-0.019208	0.691523	-0.124758	...	0.002160	0.021589	-0.707384
Additional_charges	0.005305	0.128662	0.694062	...	-0.013105	0.706033	0.025107
Item1	0.454800	-0.019010	-0.004068	...	-0.804972	-0.015568	-0.004623
Item2	0.428586	-0.020343	-0.003285	...	0.535021	0.000685	-0.000371
Item3	0.395257	-0.018942	-0.005529	...	0.191538	0.022420	0.004769
Item4	0.152012	0.051713	-0.002143	...	-0.011960	0.016999	0.002751
Item5	-0.189827	-0.060508	-0.004153	...	0.092457	0.021945	0.001413
Item6	0.410181	0.027587	-0.011216	...	0.125660	0.003042	-0.001669
Item7	0.356459	0.036128	0.005139	...	0.051216	-0.006039	0.003162
Item8	0.312456	0.026132	-0.012409	...	0.033552	-0.001471	0.004131

2.

The screeplot places the elbow at 5 principal components, however, that explains less than 60% of the data. Using the cumulative explained variance, the graph below shows that 80% of the dataset can be explained by 10 principal components. By using just 13 principal components, 90% of the data can be explained (Pramoditha, 2022).

Based on the research question, excluding too many components will not give the insights we need. Instead of keeping just 5 principal components, it makes more sense to keep 10.





3.

Reducing the number of principal components reduces the size of the data frame. A smaller data frame could be beneficial for further processing – with exploratory data analysis or creating predictive models. The extra columns that are not included as principal components could be redundant and do not necessarily add more value in further analysis.

For example, only the State and Zip code columns are necessary to explain a patient's geographic location – you can drop City, County, Longitude, Latitude, and Area.

References

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